Super Magnifier LV

Update Manual

Version 2026.05.09.0000

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Super Magnifier LV is a screen magnifier for Windows built for low-vision users. Uses the Windows Magnification API for smooth, real-time magnification compatible with DirectX games and desktop apps. Features nine colour filters, follow-cursor and follow-caret modes, multiple lens shapes, reading ruler, 5 named presets, per-monitor zoom memory, per-application auto-presets, and full WCAG 2.2 AAA accessibility.

1 What's New in v2026.05.09.0000

This release closes the outstanding WCAG 2.2 AAA compliance gaps from the 2026-05-09 audit and adds per-application automatic preset switching.

1.1 Critical Accessibility Fixes

* **CRITICAL - RadioButton / CheckBox touch targets (SC 2.5.5 AAA)** - Both ControlTemplate roots raised from MinHeight=36 to MinHeight=44. Visual indicator size unchanged; padding absorbs the delta.
* **Windows High Contrast mode (SC 1.4.8)** - HighContrast.xaml resource dictionary added (SystemColors.HighlightBrush, SystemColors.WindowTextBrush, etc.). HighContrastMonitor wired to SystemEvents.UserPreferenceChanged. Tested: High Contrast White and High Contrast Black.
* **Focus rings verified on RadioButton/CheckBox** - 2px Gold (#FFD700, 13.7:1 on black) and Cyan (#007B8A, 7.2:1 on white) applied consistently on all newly-templated controls.

1.2 New Features

* **Per-app automatic preset switching** - AppPresetMap stores foreground exe-to-preset mappings. Magnifier loads the assigned preset when a configured application gains focus.
* **Preset export / import** - Share presets as JSON independently of main settings.
* **Sleep/wake recovery** - Magnification engine auto-reinitialises after system wake.
* **What's New dialog** - Shown once per version after update to highlight changes.

1.3 Bug Fixes

* Lens position no longer drifts on secondary monitors after DPI change event.
* Off-screen recovery clamps lens to visible screen area on startup after monitor disconnect.
* Colour picker correctly saves sampled hex to clipboard on single-click.
* Colour history (last 10 colours) persists correctly across sessions.
* Snap-to-Word no longer blocks UI thread when MSAA timeout fires on unresponsive apps.

2 System Requirements

Requirement	Minimum	Recommended
OS	Windows 10 (1809+) x64	Windows 11 x64
.NET runtime	None - self-contained	None - self-contained
RAM	48 MB	128 MB+
Display	1024x600	1920x1080+
GPU	Any DirectX 9+	DirectX 11+ recommended

Note: No .NET runtime installation required. The executable is fully self-contained.

3 Update Instructions

3.1 Installer Update

* Close **Super Magnifier LV** if running (check system tray).
* Download **SuperMagnifierLV_Setup.exe** from the release page.
* Run installer. Detects existing installation and upgrades in-place.
* All settings, presets, and preferences are preserved.
* On first launch a **What's New** dialog highlights changes in this version.

3.2 Portable Update

* Download **SuperMagnifierLV_Portable_<version>.zip**.
* Extract to a temporary folder.
* Copy **SuperMagnifierLV.exe** over the existing portable executable.
* Your **Data** folder (settings, presets, logs) is untouched.

3.3 Silent Update

```
SuperMagnifierLV_Setup.exe /S
```

Installs or updates silently. Suitable for managed deployment.

3.4 Rollback

Run a previous version's installer to downgrade. Settings are preserved. Versioned builds are in the **Builds** subfolder of the project.

4 Feature Overview

Feature	Detail
Zoom range	1.25x to 20x; configurable step (0.1x/0.25x/0.5x/1x/2x).
Lens shapes	Rectangle (fixed), Circle (follows cursor), Square (follows cursor).
Follow modes	Follow cursor, Follow caret, Snap-to-Word, Edge dead zone, Smooth panning.
Colour filters	Normal, Inverted, Grayscale, High Contrast, Deuteranopia, Protanopia, Tritanopia, Warm Amber, Cool Blue.
Overlays	Reading ruler, Reticle (cross/X/circle), Cursor highlighter, Cursor finder pulse, Focus ring.
5 named presets	Zoom, lens size, colour filter, opacity, follow mode, reticle, snap-to-word. Export/import JSON.
Per-app auto-presets	Automatically switch preset when a configured application gains focus.
Per-monitor memory	Each display remembers own zoom level and lens position.
Colour picker	Click inside lens to sample any pixel; hex copied to clipboard.

Feature	Detail
Colour history	Last 10 picked colours saved as swatches.
Snapshot	Save current lens view as PNG to Pictures folder.
TTS zoom announce	Windows SAPI announces zoom level changes to screen readers.
Hotkeys	Fully remappable; all actions bindable except preset slots.
System tray	Minimise to tray; right-click for quick preset access and toggle.

5 Accessibility (WCAG 2.2 AAA)

This application targets WCAG 2.2 Level AAA across all interactive surfaces.

Criterion	Implementation
Contrast (SC 1.4.6 AAA 7:1)	All theme tokens >= 7:1. Focus rings: Gold 13.7:1, Cyan 7.2:1.
Touch targets (SC 2.5.5 AAA 44px)	Buttons 44x44px. RadioButton/CheckBox raised to 44px in this release.
Focus appearance (SC 2.4.11 AAA)	2px Gold (dark) / Cyan (light) focus ring on all controls.
No colour-only info (SC 1.4.1)	Validation hints use BrushRed plus descriptive text.
Live regions (SC 4.1.3)	Zoom readout LiveSetting=Polite. Status toasts as live regions.
High-contrast mode	HighContrast.xaml + HighContrastMonitor added in this release.
Reduce-motion (SC 2.3.3)	ClientAreaAnimation read on startup; animations skipped when off.
Keyboard navigation (SC 2.1.1)	Tab order set; Escape closes all panels and dialogs.
TTS integration	SAPI announces zoom changes; all controls have HelpText.

6 Known Issues & Notes

Privacy statement

No PII collected, no network calls, no telemetry. A formal privacy notice is planned for the next About dialog revision.

YoloMouse cursor hiding

If using a cursor-replacement tool, uncheck 'Hide cursor inside lens' in Settings.

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